

RS003 Week 8 — English Worksheet

Topic: Variable Speed — Hold Shift to Boost · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week is about how the flying **feels good**. You will think about and explain the code that makes the ship speed up while you hold **Shift**, and slowly coast to a stop with **friction**, like drifting through space. You learn about a **velocity** variable that builds up and slows down.

 Words to know: **velocity**, **acceleration**, **friction**, **max speed**, **Shift key**

1 · Predict

Read each piece of code. Just by reading it, write what you think will happen.

```
velocity_x = 0
ACCEL = 0.5
velocity_x += ACCEL
ship_x = ship_x + int(velocity_x)
```

velocity_x keeps growing while you hold a key. What happens to the ship's movement — does it move at a steady speed or get faster?


```
if not keys[pygame.K_LEFT] and not keys[pygame.K_RIGHT]:
    velocity_x = velocity_x * FRICTION    # FRICTION = 0.85
```

When you let go of both keys, the velocity is multiplied by 0.85 every loop. What does that do over time?


```
max_v = MAX_SPEED_FAST if keys[pygame.K_LSHIFT] else MAX_SPEED
```

What does holding Shift change about the top speed?


```
if velocity_x > max_v:  
    velocity_x = max_v  
if velocity_x < -max_v:  
    velocity_x = -max_v
```

If `velocity_x` tries to grow past `max_v`, what do these lines do to it?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Write the fixed code, then explain why the original was wrong.

Bug A — The ship should coast to a stop when you let go, but instead it keeps sliding forever.

```
if keys[pygame.K_LEFT]:
    velocity_x -= ACCEL
if keys[pygame.K_RIGHT]:
    velocity_x += ACCEL
ship_x += int(velocity_x)
```

Hint: nothing slows the velocity down when no key is held.

Write the fixed code (add the friction step):

Why was it wrong? Why does your fix work?

Bug B — The ship is supposed to have a top speed, but holding a key for a long time makes it shoot across the screen far too fast.

```
if keys[pygame.K_RIGHT]:
    velocity_x += ACCEL
ship_x += int(velocity_x)
```

Hint: the velocity is never capped at a maximum.

Write the fixed code (add the speed limit):

Why was it wrong? Why does your fix work?

Bug C — The program crashes because Pygame is given a position with a decimal point.

```
ship_x = ship_x + velocity_x
pygame.draw.rect(screen, CYAN, (ship_x - 20, ship_y, 40, 50))
```

Hint: velocity is a decimal; screen positions must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this full program. It makes the ship speed up, coast, and boost. Answer the questions by reading only — you do not need to run it.

```

import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

ship_x = CENTER_X
ship_y = HEIGHT - 100
velocity_x = 0
ACCEL = 0.5
MAX_SPEED = 8
MAX_SPEED_FAST = 16
FRICTION = 0.85

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    max_v = MAX_SPEED_FAST if (keys[pygame.K_LSHIFT] or keys[pygame.K_RSHIFT]) else MAX_SPEED

    if keys[pygame.K_LEFT]:
        velocity_x -= ACCEL
    if keys[pygame.K_RIGHT]:
        velocity_x += ACCEL

    if velocity_x > max_v:
        velocity_x = max_v
    if velocity_x < -max_v:
        velocity_x = -max_v

    if not keys[pygame.K_LEFT] and not keys[pygame.K_RIGHT]:
        velocity_x = velocity_x * FRICTION

    ship_x = ship_x + int(velocity_x)

    screen.fill((10, 12, 40))
    pygame.draw.rect(screen, (80, 220, 255), (ship_x - 20, ship_y, 40, 50))
    pygame.draw.polygon(screen, (255, 255, 255), [(ship_x, ship_y - 20), (ship_x - 20, ship_y), (ship_x + 20, ship_y)])

```

```
pygame.display.flip()
clock.tick(60)

pygame.quit()
```

Which line makes the ship speed up when you press RIGHT, and how does it change `velocity_x` ?

What does the line `max_v = MAX_SPEED_FAST if ... else MAX_SPEED` decide, and what makes it pick the fast number?

The line `velocity_x = velocity_x * FRICTION` only runs when no arrow key is held. Why is that the right time to slow the ship down?

Why does the code use `int(velocity_x)` instead of just `velocity_x` when changing `ship_x` ?

velocity_x can become a negative number. What does a negative velocity make the ship do?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote your own ship-with-feel code. Now explain it. Record a short phone video. You may show your ship running while you talk. Try to use these words: **velocity**, **acceleration**, **friction**, **max speed**, **Shift**.

1. Show your ship flying slowly, then hold Shift and fly fast.
2. Let go of the keys and show the ship coasting to a stop.
3. Point to your friction line and say what 0.85 does.
4. Explain how your code stops the ship going past the max speed.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.